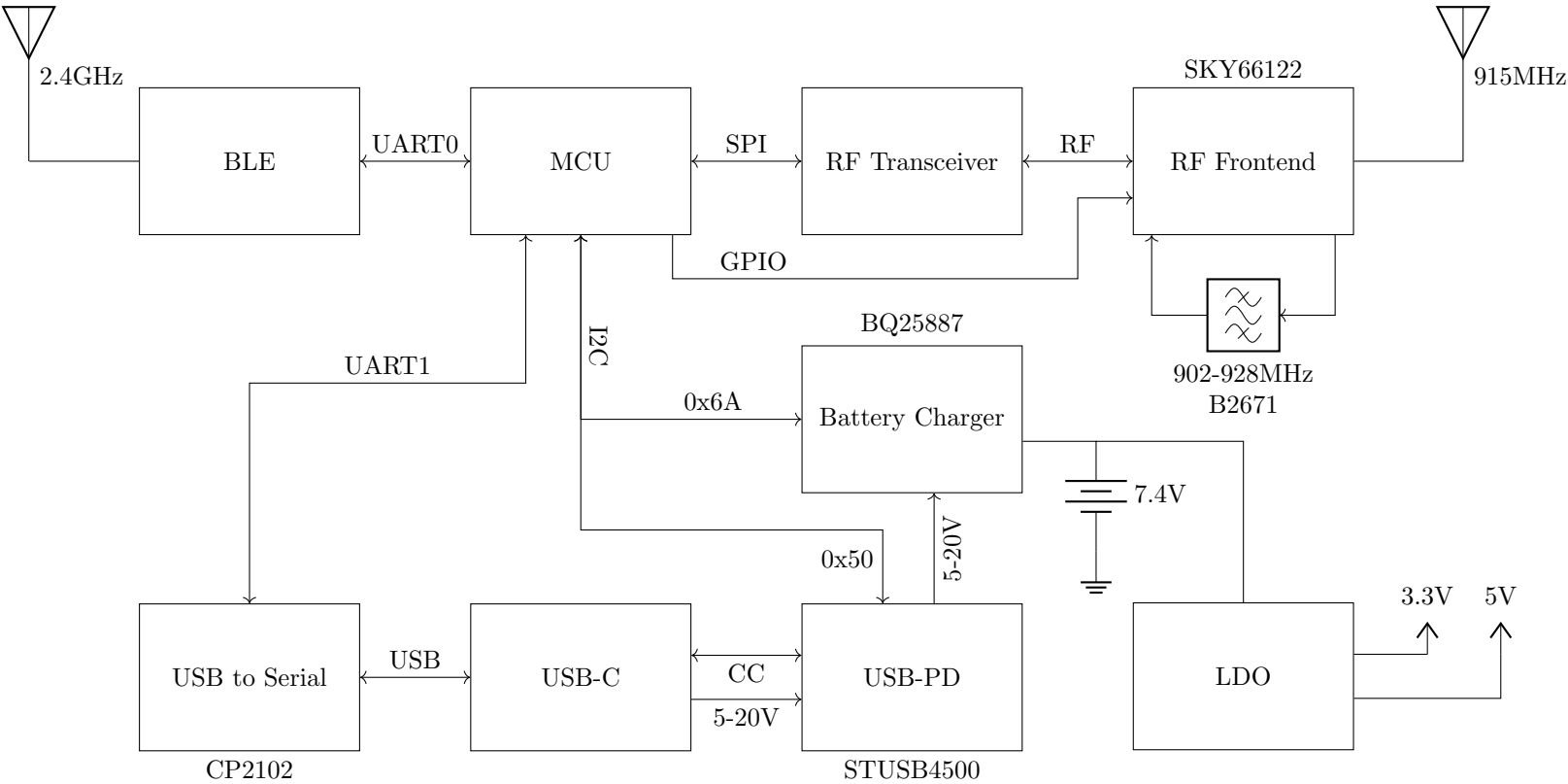
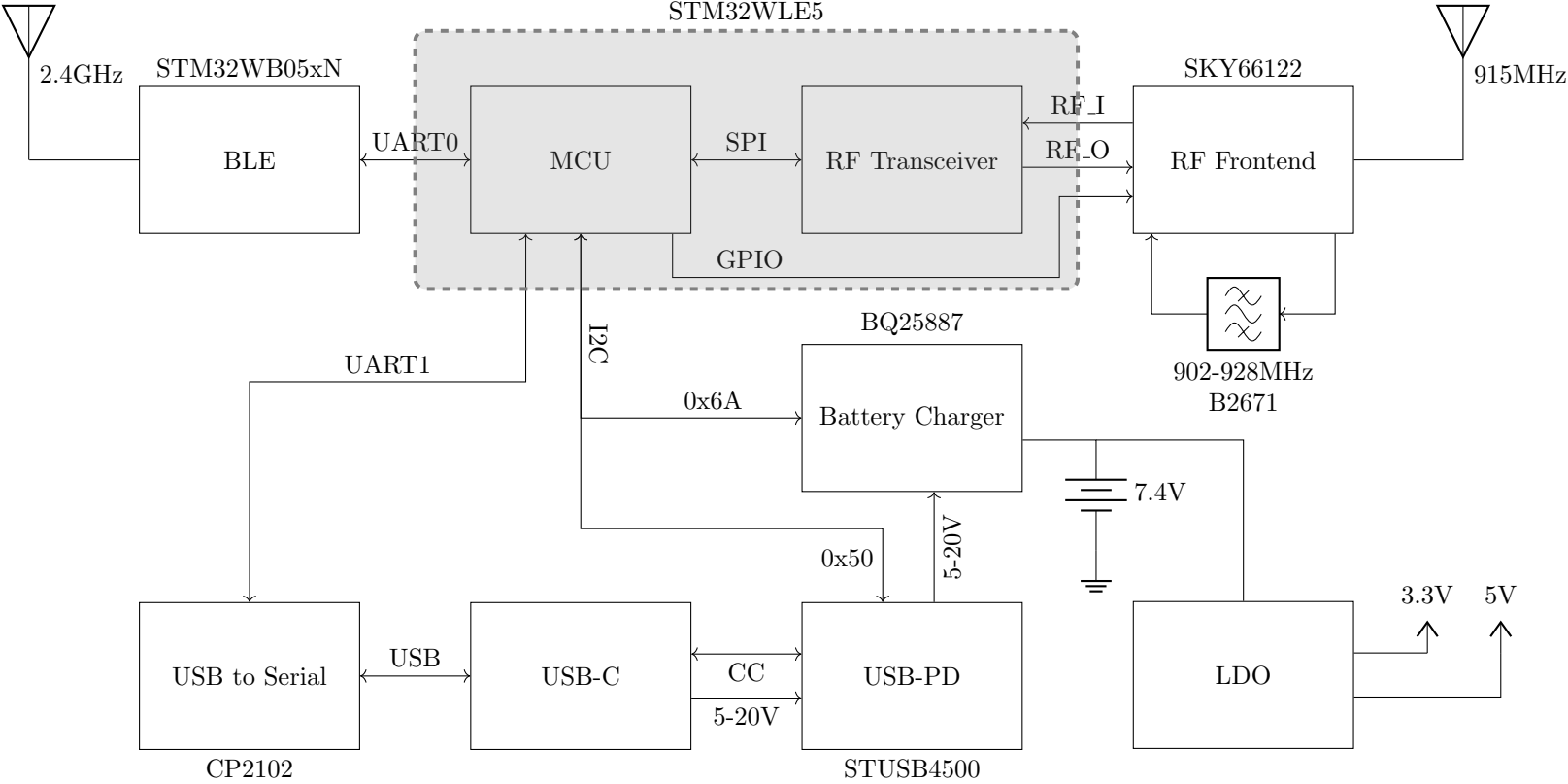


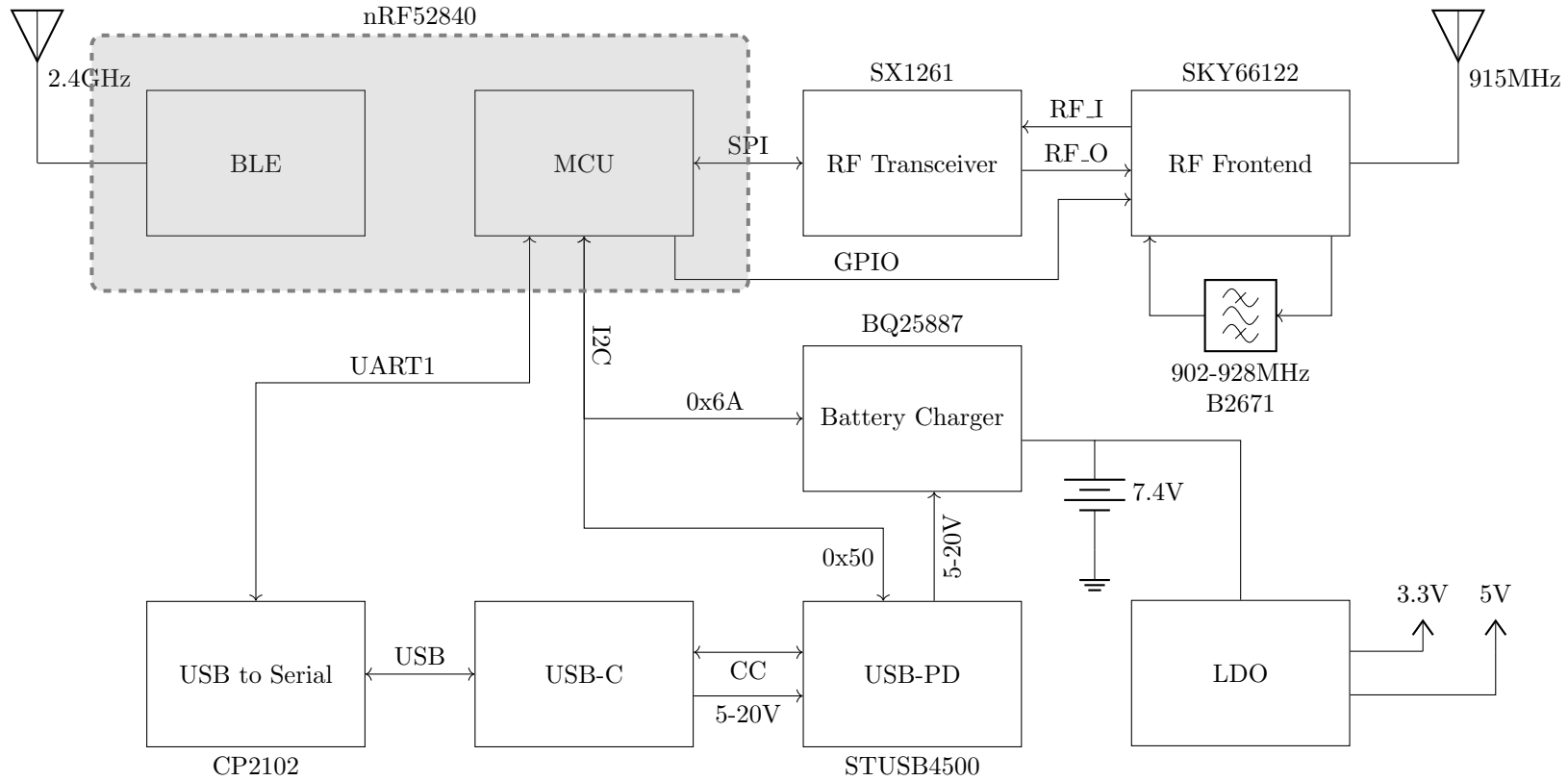
General Overview Block Diagram



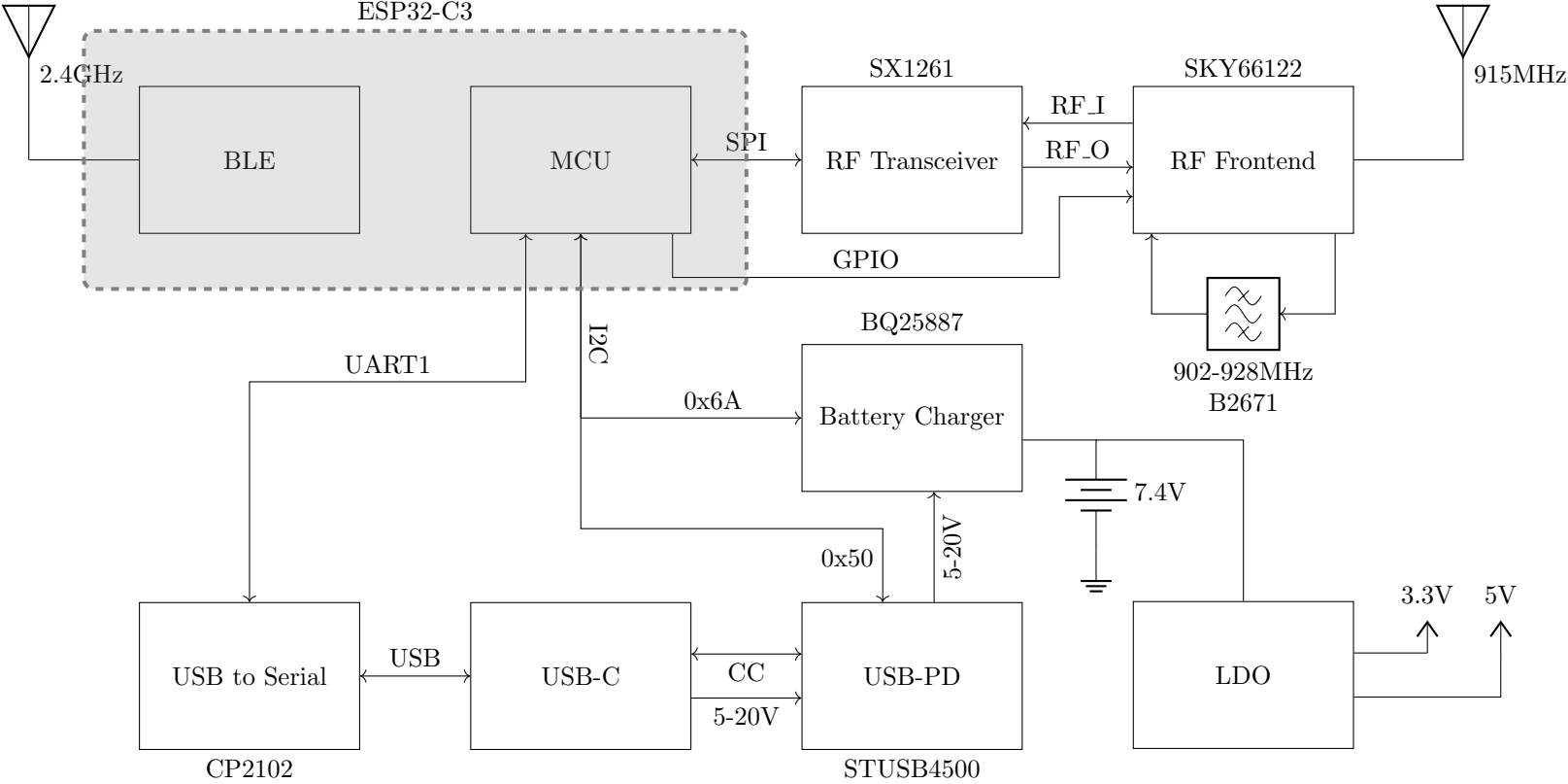
Component Selection Option A (STM32)



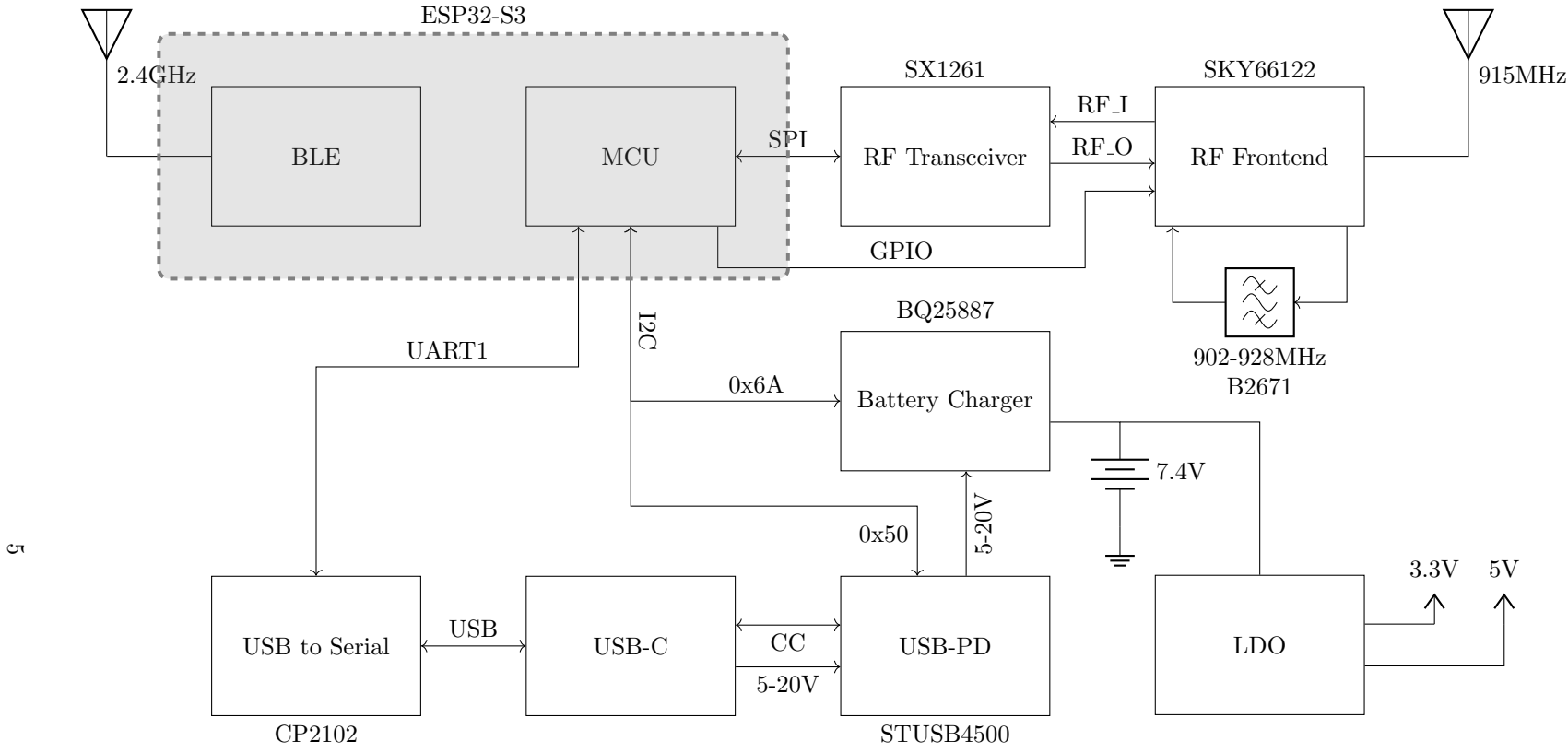
Component Selection Option B (nRF52840)



Component Selection Option C-1 (ESP32C3)



Component Selection Option C-2 (ESP32S3)



Option A

Component	Part Number	Peak I (mA)	Continuous I (mA)	Sleep I (μ A) ¹
BLE	STM32WB05xN	5.6	4.950	0.8 ²
MCU	STM32WLE5	5.70 ³	0.011 ⁴	0.935
RF Transceiver		12 ⁵	5.3 ⁶	0.6 ⁷
RF Frontend	SKY66122	640	6.5	0.05
Total		663.3 mA	16.76 mA	2.385 μA

¹The sleep mode current is with the LoRA RX off (the receiver has no passive receive sleep mode. But with enabled BLE passive RX. There is a Channel Activity Detection (CAD) module implemented in the SX12xx series devices, but the use thereof is not implemented by *meshtastic* for sleeping)

²Wake on BLE Mode

³LDO Mode, Range 1

⁴LPSleep Mode, Flash OFF

⁵TX Current Draw

⁶RX Current Draw

⁷OFF Current draw (config retained)

Option B

Component	Part Number	Peak I (mA)	Continuous I (mA)	Sleep I (μ A)
BLE	nRF52840	6.4	6.26	2.35 ¹
MCU		6.3 ²	0.01737 ³	0.935
RF Transceiver	SX1261	12	5.3	0.6
RF Frontend	SKY66122	640	6.5	0.05
Total		664.7 mA	18.08 mA	3.935 μA

¹System ON, full 256 kB RAM retention, wake on any event

²LDO Mode

³System ON, full 256 kB RAM retention, wake on GPIOTE input (event mode)

Option C-1

Component	Part Number	Peak I (mA)	Continuous I (mA)	Sleep I (μ A)
BLE	ESP32-C3			
MCU				
RF Transceiver	SX1261	12	5.3	0.6
RF Frontend	SKY66122	640	6.5	0.05
Total		664.7 mA	18.08 mA	3.935 μA

Option C-2

Component	Part Number	Peak I (mA)	Continuous I (mA)	Sleep I (μ A)
BLE	ESP32-C3			
MCU				
RF Transceiver	SX1261	12	5.3	0.6
RF Frontend	SKY66122	640	6.5	0.05
Total		664.7 mA	18.08 mA	3.935 μA

1 LDO Selection